TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 2

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

exemplar paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Past Paper 2 (Section A Non-Calculator & Section B Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 35 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section A	15		
Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 9.028 – 0.17

Step 1: Pad decimals to equal places: 14.508 - 5.650

14.508 - 5.650 = 8.858

Answer 8.858 ✓ *[1 mark]*

[1]

2. Work out 24.108 ÷ 4

Step 1: Short division: 24.108 ÷ 4

24 ÷ 4 = 6; .1 ÷ 4 = 0 r 1; 10 ÷ 4 = 2 r 2; 28 ÷ 4 = 7 --> 6.027

Answer 6.027 ✓ *[1 mark]*

[1]

Past paper

Questions continue on the following page

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3. 50 people are asked what pets they own.

The results are shown below:

	Rabbit	Cat
Dog	16	24
Fish	2	8

One person is chosen at random.

What is the probability that the person chosen has a dog and a cat?

Give your answer as a decimal.

Step 1: Express probability as a fraction: P = 24 / 50

Step 2: Convert to decimal: 24 ÷ 50 = 48 ÷ 100 = 0.48 [M1]

Answer 0.48 ✓ [A1]

[2]

4. The mass of a sphere is 1800 grams (g).

The volume of the sphere is 450 cm^3

Work out the density of the sphere.

Step 1: Density formula: Density = Mass ÷ Volume

Density = 1800 g ÷ 450 cm^3 = 4 g/cm^3 [M1]

Answer 4 ✓ [A1]

[2]

Past paper

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5. Jesse is the manager of an aquarium.

The table below shows how many people visited the aquarium over 3 years:

Year	Number of people
2019	568 750
2020	128 528
2021	482 935

Jesse thinks that more than 1 200 000 people visited the aquarium during the 3 years.

Is Jesse correct?

Show how you decide.

Step 1: Total visitors = 568 750 + 128 528 + 482 935 = 1 180 213 [M1]

Step 2: Compare with 1 200 000: 1 180 213 < 1 200 000, so Jesse is incorrect.

Answer No (1 180 213 < 1 200 000) [A1]

[2]

6. Work out 1.324 + 2.206 + 14.75

Step 1: Align by decimal point: 1.324 + 2.206 + 14.750

Sum = 18.280 --> 18.28 [M1]

Answer 18.28 [A1]

[2]

Questions continue on the following page

Past paper

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7. Work out:

$$\frac{3}{20} + \frac{17}{15}$$

Give your answer as an improper fraction.

Step 1: Find LCM of denominators (20 and 15): LCM = 60

3/20 = 9/60 and 17/15 = 68/60 [M1]

Step 2: Add numerators: 9/60 + 68/60 = 77 / 60 (or 1 17/60)

Answer	77 / 60 ✓ [A1]

[2]

8. Riley's annual salary is £32 000 before tax.

Riley knows that:

- £12 570 will be tax free
- 20% tax will be deducted from the amount above £12 570

How much will the annual salary be after tax has been deducted?

Step 1: Tax-free allowance = £12 570. Taxable pay = £32 000 - £12 570 = £19 430 [M1]

Step 2: Income tax at 20%: £19 430 × 0.20 = £3886 [M1]

Step 3: Net salary = £32 000 - £3886 = £28 114

Answer	£ 28 114 ✓ [A1]

[3]

End of Section A.

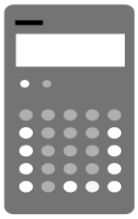
Past paper

Section B begins on Page 6.

Past paper

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Section B: Calculator



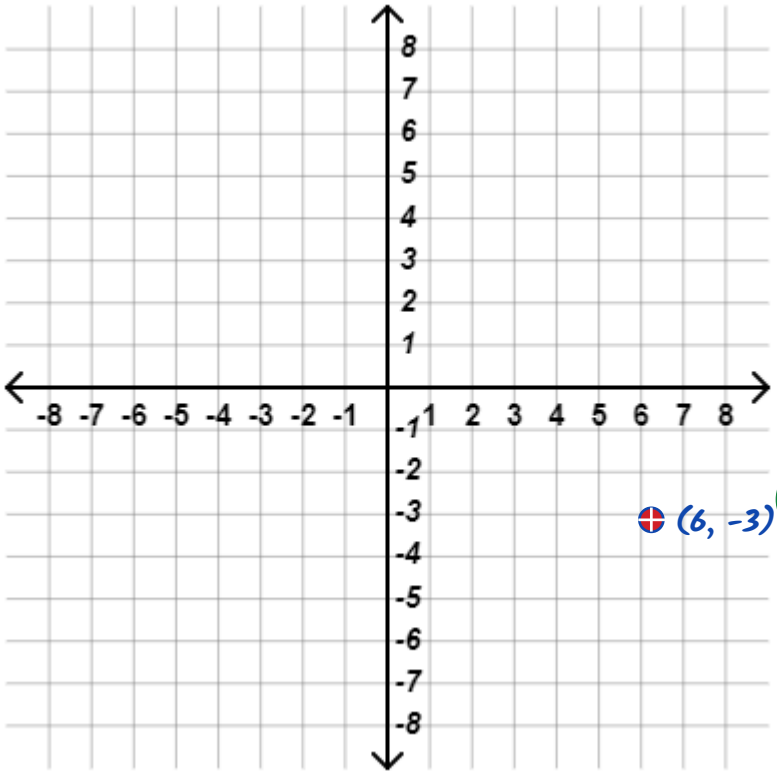
There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. Write 13 000 000 000 in words.

Answer	Thirteen billi ✓ [1 mark]

[1]

2. Plot the point (6, –3) on this coordinate grid:



[1 mark]

[1]

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3. At football training, a footballer has 56 shots at goal and scores 42 times.

The football coach thinks that the footballer scored $\frac{7}{8}$ of the shots.

Has the coach worked out the correct fraction?

Show how you decide.

Step 1: Simplify fraction scored: 42 ÷ 56 = 3/4 = 6/8

Step 2: Compare with coach's claim: 7/8 of 56 = 49 shots (not 42) [M1]

6/8 ≠ 7/8, so the coach is incorrect.

Answer	No (fraction is 3/4 or 6/8) [A1]

[2]

4. Work out:

$$\left(\frac{25+5\times3}{10^2}\right)$$

Give your answer as a decimal.

Step 1: Apply BIDMAS - evaluate brackets/fractions first:

Calculation result = 0.4 [M1]

Answer	0.4 [A1]

[2]

Questions continue on the following page

Past paper

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5. 3480 people visited a museum in July.

5150 people visited the same museum in August.

Calculate the percentage change in the number of people from July to August.

Step 1: Increase in visitors = 5150 - 3480 = 1670

Step 2: Percentage change = (1670 ÷ 3480) × 100 = 47.98...% [M1]

Rounded: 48% (or 48.0%)

Answer	48 ✓ [A1] %

[2]

6. Leslie is planning a road trip.

The scale on the map is 1 : 75 000

The distance Leslie wants to travel represents 11 centimetres (cm) on the map.

Leslie calculates this to be 82.5 kilometres (km).

Is Leslie correct?

Show how you decide.

Step 1: Actual distance in cm = 11 cm × 75 000 = 825 000 cm [M1]

Step 2: Convert cm to km: 825 000 ÷ 100 000 = 8.25 km

8.25 km ≠ 82.5 km (Leslie is out by a factor of 10).

Answer	No (actual distance is 8.25 km) ✓ [A1]

[2]

Past paper

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7. Nicky's homemade pillowcases are made using a piece of fabric two metres square.

Each time a pillowcase is made, 0.135 square metres of fabric is wasted.

Nicky wants to **estimate** what percentage of the total fabric is wasted.

Nicky rounds 0.135 to one decimal place.

Using the rounded value, what percentage should Nicky get?

Show your working.

Step 1: Round waste 0.135 m^2 to 1 decimal place: 0.135 --> 0.1 m^2 [M1]

Step 2: Percentage of 2 m^2 fabric wasted = (0.1 ÷ 2) × 100 = 5% [M1]

Answer 5 ✓ [A1]	%

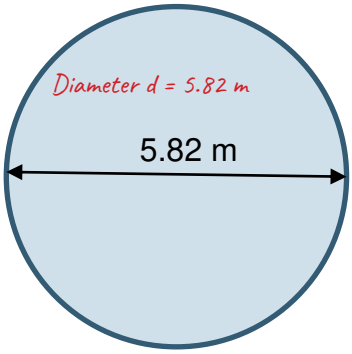
[3]

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Questions continue on the following page

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8. Charlie wants to put a safety rope around this pond:



Not drawn to scale

Charlie thinks the total length of rope needed will be more than 18.5 metres (m).

Is Charlie correct?

Show how you decide.

Use $\pi = 3.14$

Step 1: Circumference formula: C = pi × d

C = 3.14 × 5.82 m = 18.2748 m [M1]

Step 2: Compare with rope: 18.27 m < 18.5 m, so rope is long enough.

Answer	No (pond is 18.27 m < 18. ✓ [A1]

[3]

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9. Harper paid £12 500 into a new savings account 2 years ago.

The savings account paid 3% compound interest per year.

Harper wants to buy a new car for £18 750

Harper puts all the money in the savings account towards the new car.

How much more money is needed to pay for the car?

Step 1: Calculate 3% compound interest for 2 years:

Year 1: £12 500 × 1.03 = £12 875 | Year 2: £12 875 × 1.03 = £13 261.25 [M1]

Step 2: Money needed = £18 750 − £13 261.25 = £5488.75 [M1]

Answer £ 5488.7 [A1]

[3]

Past paper

Questions continue on the following page

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10. Alex needs to know what time to leave home to get to a dentist appointment by 10:40 am.

Alex can cycle at an average speed of 12 miles per hour.

The dental surgery is 15 miles from Alex's house.

What is the **latest** time Alex can leave home to cycle to the appointment?

Step 1: Time = Distance ÷ Speed = 15 miles ÷ 12 mph = 1.25 hours [M1]

1.25 hours = 1 hour + (0.25 × 60) mins = 1 hour 15 minutes

Step 2: Latest departure = 10:40 am - 1 hr 15 mins = 9.25 am [M1]

Answer 9:25 [A1] **am**

[3]

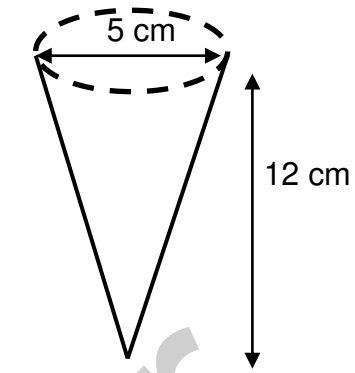
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11. Taylor is going to fill a cone with strawberry and vanilla ice-creams.

Taylor will use strawberry and vanilla in the ratio 2 : 3

How much of each flavour will Taylor need?



Not drawn to scale

Use:

$$v = r^2h$$

Where:
v = volume of cone
r = radius of cone
h = height of cone

Step 1: Cone volume formula: $V = r^2 \times h$

Radius $r = 5 \div 2 = 2.5$ cm --> $V = 2.5^2 \times 12 = 6.25 \times 12 =$ [M1] 75 cm³

Step 2: Ratio Strawberry : Vanilla = 2 : 3 (Total = 5 parts)

1 part = 75 cm³ $\div 5 = 15$ cm³ [M1]

Strawberry = $2 \times 15 = 30$ cm³ | Vanilla = $3 \times 15 = 45$ cm³

Strawberry	30 ✓	**cm³**
Vanilla	45 ✓ [A1]	**cm³**

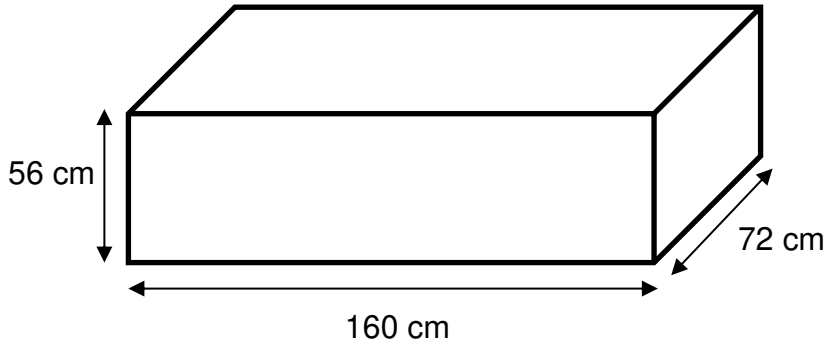
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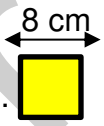
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12. Jamie needs to cover this cuboid chest in multi-coloured stickers for an art project:



Not drawn to scale



The stickers are in the shape of a square with a side length of 8 cm.

To work out how many stickers are needed, Jamie uses this formula:

$$\text{Total number of stickers} = \frac{\text{Total surface area of chest}}{\text{Area of one sticker}}$$

How many stickers does Jamie need?

Step 1: Surface area of cuboid = 2(lw + lh + wh)

= 2[(160 × 72) + (160 × 56) + (72 × 56)] = 2(11520 + 8960 [M1] 4032) = 49 024 cm²

Step 2: Area of 1 square sticker = 8 cm × 8 cm = 64 cm²

Step 3: Stickers needed = 49 024 ÷ 64 = 766 stickers [M1]

Answer 766 [A1]

[4]

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13. Drew is a gardener.

This table shows how much other local gardeners charge per hour:

Amount (A) charged per hour (£)	Number of gardeners
20 < A ≤ 30	2
30 < A ≤ 40	6
40 < A ≤ 50	8
50 < A ≤ 60	4
Total	20

Drew wants to charge an hourly rate equal to the estimated mean amount per hour.

How much should Drew charge per hour?

Step 1: Find midpoints (x) for each hourly rate interval:

20<A≤30: x=25 (f=2, fx=50) | 30<A≤40: x=35 (f=6, fx=210)

40<A≤50: x=45 (f=8, fx=360) | 50<A≤60: x=55 (f=4, fx=220) [M1]

Step 2: Total fx = 50 + 210 + 360 + 220 = 840; Total gardeners = 20

Step 3: Estimated Mean = 840 ÷ 20 = £42 per hour [M1]

Answer £ 42 [A1]

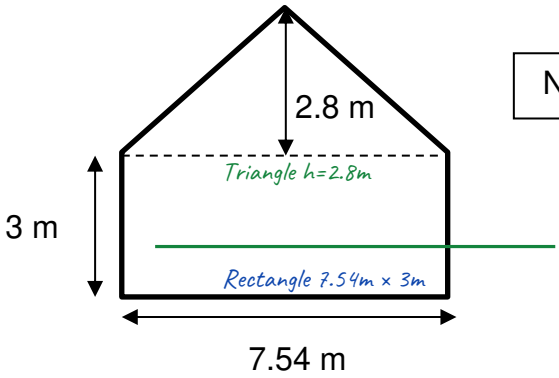
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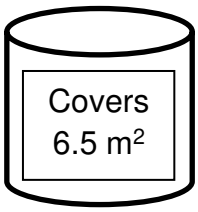
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14. Ashley wants to paint the wall shown:



Not drawn to scale

Ashley finds this tin of paint online:



How many tins of paint will Ashley need to buy to paint the wall?

Show your working.

Step 1: Area of top triangle = 0.5 × base × height = 0.5 × 7.54 × 2.8 = 10.556 m² [M1]

Step 2: Area of bottom rectangle = 7.54 m × 3 m = 22.62 m²

Step 3: Total wall area = 10.556 + 22.62 = 33.176 m² [M1]

Step 4: Tins required = 33.176 ÷ 6.5 = 5.104 tins

Round up to whole tins: 5.104 --> 6 tins

Answer	6 ✓ [A1]	**tins**

[5]

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15. Quinn wants to buy some rope lights to put around a 164 foot section of a garden.

Quinn finds these rope lights online:

Rope light	A	B	C	D	E
Price per metre	£0.60	£1.99	£1.89	£0.75	£0.50

Quinn wants to buy rope lights that are 8% longer than the section of the garden.

Quinn buys the rope lights with the median price.

How much will Quinn pay in total?

Use 1 foot = 0.305 metres

Step 1: Order prices: £0.50, £0.60, £0.75, £1.89, £1.99 --> Median = £0.75 [M1]

Step 2: Convert 164 ft to metres (1 ft = 0.305 m): 164 × 0.305 = 50.02 m [M1]

Step 3: Add 8% extra length: 50.02 m × 1.08 = 54.0216 m (or 50 × 1.08 = 54 m) [M1]

Step 4: Total cost = 54 m × £0.75 = £40.50 (or 54.0216 × 0.75 = £40.52)

Answer £ 40.50 [A1]

[6]

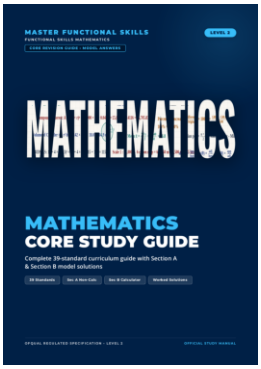
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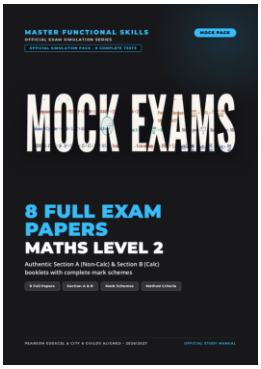


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